Decarbonising today's power system

Expansion at European scale

Reading. *Energy System Models* (ESM below), section 8, and for part 3 section 5 too.

The Lab and the Explorer. This worksheet holds the questions of lecture 7 in the Model Lab, where you answer them a part at a time. The figures they need are in the Model Explorer: every scenario of the section 8 model, solved in advance and drawn figure by figure, with a description of the model under "About this model".

The Model Lab	*championape-model-lab.share.connect.posit.cloud*
The Model Explorer	*championape.github.io/EnergyEconomics-explorer*

Each question names the view it needs: the Explorer's tab and figure, the scenario, and the scenario it is compared with where there is one. The link opens that view.

Part 1 · ESM §8.2 30 min

The baseline: prices and the last quarter

Section 8 explores how to reduce emissions in the current system. We refer to both emission budgets and carbon prices/taxes: they are not separate instruments in this model, the carbon price/tax is simply the shadow price of the emission budget. Emission cuts are simply the inverse to budgets: A cut of 99 % is a budget of 1 %.

1. *What the cut changes.*

> **In the Explorer:** Overview › *Energy balance: all twelve zones*, *Baseline, budget 1 %* against *Baseline, no budget*: generation by technology at a cut of 99 % and, faded, with no cut.

(i) From figures 8.1 and 8.2, compare how the system changes from no cut to a cut of 75 % with how it changes from 75 % to 99 %. *at most 150 words*

(ii) Why does the price distribution change between no cut and a cut of 99 %?

> **In the Explorer:** Prices and margins › *The price-duration curve*, all twelve zones, *Baseline, budget 1 %* against *Baseline, no budget*: the system's price duration curve at a cut of 99 % and, dashed, with no cut.

(a) Storage sets the price at its own running cost, close to zero, so the more storage is built, the more hours are near zero, while the carbon price makes the remaining dear hours dearer.

(b) The zones trade less with each other, so each zone's price follows its own wind and sun more closely and swings further, towards zero and towards scarcity.

(c) Curtailed wind and solar set the price near zero in more hours, and dearer fuel (coal paying for its carbon, biomethane) makes the hours set by plants burning fuel dearer; storage carries both levels between hours.

(d) The flexible demand block steps back more often, the price falls close to zero in the hours it steps back, and the other hours are dearer.

2. *What covers the last quarter.* Four margins compete for the hours with the highest residual demand: storage, the flexible demand block, backup capacity and the interconnectors (section 8.2).

> **In the Explorer:** Prices and margins › *What one more MW of storage would earn*, *Baseline, budget 1 %* against *Baseline, no budget*: what one more MW of storage would earn over its annual cost, at a cut of 99 % and, dashed, with no cut.

(i) What do the three curves of figure 8.3(b) say about the value of storage that holds days or weeks as the cut deepens? Use the Explorer figure below to explain why, and why such storage would still not pay for itself. A MW of power costs 9.7 k€ a year, a MWh of energy capacity 6.4 k€. *at most 150 words*

> **In the Explorer:** Prices and margins › *How often storage would cycle, and what each MWh would earn*, the same two scenarios: how often one more MW of storage would cycle (above) and what it would earn per MWh delivered (below), at a cut of 99 % and, dashed, with no cut.

(ii) The flexible demand block steps back whenever the price is above 100 €/MWh: in about 400 hours a year at a cut of 75 % (carbon price 66 €/t) and in about 3,600 at 90 % (205 €/t). A new combined-cycle plant runs at 34.3 €/MWh before carbon and emits 0.33 t/MWh. What explains the jump?

> **In the Explorer:** Sweeps: *Flexible demand steps back* against *CO2 budget, all twelve zones*: the hours in which the flexible demand block steps back, against the budget, one line per scenario.

(a) The budget forbids the gas plants to run at a cut of 90 %, so the block steps back in all the hours they covered at 75 %.
(b) Between the two cuts the carbon price passes about 200 €/t, above which even the best gas plant on natural gas costs more to run than the block will pay, so the block steps back in its place.
(c) At a cut of 90 % the batteries run empty for most of the winter, so only the block is left to serve the evening peak.
(d) The block's willingness to pay falls as the carbon price rises, since the industry behind it pays the carbon price on its own fuel.

If you finish early

The value of an interconnector.

In the Explorer: Grid, emissions and money › *Interconnectors bought*, *Baseline, budget 1 %* against *Baseline, no budget*: the interconnectors bought at a cut of 99 % and, faded, with no cut, against the most each border may take.

In section 8's notation, a candidate interconnector k carries $f_{k,h}$ from zone a_k to zone b_k, where $\eta_k f_{k,h}$ arrives. Its capacity $\bar{f}_k$ costs F_k a MW a year and may be bought up to a ceiling:

$$0 \le f_{k,h} \le \bar{f}_k \quad (\underline{\omega}_{k,h}, \overline{\omega}_{k,h}), \qquad 0 \le \bar{f}_k \le \bar{f}_k^{\max} \quad (\underline{\beta}_k, \overline{\beta}_k).$$

The prices at the two ends are $\lambda_{a_k,h}$ and $\lambda_{b_k,h}$.

(i) Derive Result 7 for a candidate interconnector k. Use the first-order conditions for its flow $f_{k,h}$ and its capacity $\bar{f}_k$, and complementary slackness on $0 \le f_{k,h} \le \bar{f}_k$, to show that

$$\overline{\beta}_k - \underline{\beta}_k = \sum_h \max\left(\eta_k \lambda_{b_k,h} - \lambda_{a_k,h},\ 0\right) - F_k.$$

Say in words what the sum on the right-hand side is. *at most 250 words*

(ii) Read the condition for a cable bought below its ceiling: what does one more MW of it earn, in which hours, and why is that also the system cost it saves? Why would an owner paid only that rent buy less of the cable than the planner? *at most 150 words*

(iii) At a cut of 99 %, one more MW of cable between Germany and southern Norway would earn about 290 k€ a year in congestion rent, more than on any border where the model buys cable. Yet it buys none there. Why?

- (a) The border is congested in too few hours for a new cable to earn its cost.
- (b) A new cable would lower the rents on the other borders, so the planner leaves it out.
- (c) The rent would go to the two transmission operators rather than to the system, so the planner does not count it.
- (d) A MW of that long subsea cable costs about 380 k€ a year, more than it would earn, so the condition holds at its lower bound.

Electric vehicles: half a battery and a whole one

1. *What the cars do to prices.*

> **In the Explorer:** Prices and margins › *The price-duration curve*, all twelve zones, *Vehicle-to-grid, budget 1 %* against *Baseline, budget 1 %*: the system's price duration curve at a cut of 99 %, vehicle-to-grid and, dashed, the baseline.

(i) From figure 8.4, compare and explain how smart charging and vehicle-to-grid change the system's prices against the baseline. *at most 150 words*

(ii) With vehicle-to-grid the cars set the price for more than half of the year's demand. What fixes the price in an hour in which the cars are at the margin?

> **In the Explorer:** Prices and margins › *What sets the price*, *Vehicle-to-grid, budget 1 %* against *Smart charging, budget 1 %*: what sets the price at a cut of 99 %, vehicle-to-grid and, faded, smart charging.

(a) What a MWh in the cars' batteries is worth: like a battery, the cars pass on the price of the hours they trade with, net of their losses.

(b) Their running cost, close to zero, so the price falls towards zero whenever the cars are at the margin.

(c) The cost of the cars' batteries and chargers, which the price must cover in the hours the cars sell.

(d) The driving: the cars must be charged whatever the price, so the price rises until they are.

2. *Cars against batteries.*

> **In the Explorer:** Prices and margins › *What covers the scarcest hours*, *Vehicle-to-grid, budget 1 %* against *Baseline, budget 1 %*: what covers the scarcest tenth of the year at a cut of 99 %.

(i) In figure 8.5(b), one more MW of eight-hour storage still earns exactly its cost with smart charging, but not with vehicle-to-grid. Explain why with the stationary battery's investment condition, and what it means for the cheapest hours of figure 8.4. *at most 120 words*

> **In the Explorer:** Prices and margins › *What one more MW of storage would earn*, *Vehicle-to-grid, budget 1 %* against *Smart charging, budget 1 %*: what one more MW of storage would earn over its annual cost at a cut of 99 %, vehicle-to-grid and, dashed, smart charging.

(ii) With vehicle-to-grid, one more MW of eight-hour storage no longer pays, while one more MW of week-long storage earns almost as much as in the baseline. Why do the cars take so little from long storage?

(a) The cars' batteries cost the power system nothing, so the cars are used before any storage that must be paid for, whatever its duration.

(b) Vehicle-to-grid runs mainly in summer, when the cars stand still for longer, while the long stretches of high prices come in winter.

(c) Like the batteries they replace, the cars trade on the daily spread: half their batteries, drained by the driving and kept above a floor for the morning's trips, cannot carry energy across a calm week.

(d) With vehicle-to-grid the carbon price is lower, which leaves the spread between a windy week and a calm one where it was.

Hydrogen: a flexible load, not a storage

1. *The hydrogen sector, part by part.*

> **In the Explorer:** Hydrogen › *Hydrogen supply and use: all twelve zones*, *Reference: hydrogen sector, budget 1 %*: a year's hydrogen in all zones, reformed, imported, electrolysed and used.

(i) Compare the hydrogen sector of section 8.4 with the heat sector of section 5: what role does each part of the hydrogen sector play, which part of the heat sector plays the same role, and where does the comparison break down? *at most 180 words*

(ii) What would make hydrogen pay as seasonal electricity storage in the model? Select all that apply.

- (a) If the carbon price were higher, with biomethane as plentiful and cheap as before.
- (b) If biomethane, the gas turbines' carbon-free fuel, were not available.
- (c) If there were no industrial demand for hydrogen, so that all of it were free for the turbines.
- (d) If hydrogen in storage were cheaper than biomethane in the hours when power is dear.
- (e) If there were no hydrogen imports, so that more hydrogen were made from electricity.
- (f) If biomethane were much dearer.

2. *Electrolysers as a flexible load.* In the hydrogen scenario at a cut of 99 %, the reformers burn biomethane only. Biomethane costs 62 €/MWh, and a reformer turns 72 % of it into hydrogen; an electrolyser turns 70 % of the electricity it draws into hydrogen.

> **In the Explorer:** Hydrogen › *Electrolysers against the electricity price: DE, Germany*, *Reference: hydrogen sector, budget 1 %*: the electricity price against the share of the electrolysers' capacity that ran, one point per segment of the year.

(i) In Germany a reformer burning biomethane is the marginal source of hydrogen in most hours. Derive the electricity price below which an electrolyser runs flat out, and say what sets the price when it runs part-loaded. Then explain why, against smart charging, the hydrogen scenario has fewer hours near zero yet a higher average price. *at most 120 words*

> **In the Explorer:** Prices and margins › *The price-duration curve*, all twelve zones, *Reference: hydrogen sector, budget 1 %* against *Smart charging, budget 1 %*: the hydrogen scenario and, dashed, smart charging. Their budgets differ in tonnes.

(ii) The electrolysers earn exactly their fixed cost, and in many of the hours they run, they run part-loaded and set the electricity price. Where does the money that pays for their capital come from?

- (a) From the hours in which they set the price: as the marginal consumer they choose what they pay, and pay less than their hydrogen is worth.
- (b) From the hours in which electricity is cheaper than the hydrogen it makes and they run flat out; in the hours in which they set the price, they earn nothing.
- (c) From hydrogen made in summer, when electricity is cheap, stored in salt caverns and sold to industry in winter at a higher price.
- (d) From the scarcest hours, in which the hydrogen price is highest and the electrolysers run flat out before the reformers do.